

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWI")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,          TruncNormal,              0.1,          0.2,          0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,          0.8,          0.1,          0.1, 2);
Distrib (kap,          Uniform,                  0.1,          0.9);
Distrib (Ehp,          TruncNormal_cv,          100,          0.2,          40, 300);
Distrib (E_coc,        TruncNormal_cv,          30,          0.2,          1, 300);
Distrib (rOM_ClxHorse, LogUniform,              0.00001,      0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihoood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihoood, NbIter)

# ./mcsim.DEBcaliSWI DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	140.081000	0.64196300	0.100242000	0.899184000	198.68600	63.683700
2.5%	131.855000	0.47280600	0.100086000	0.882338000	167.24800	56.591400
97.5%	158.194850	0.74748795	0.115824500	0.899882000	232.86900	73.979500
Min	87.590600	0.09270370	0.100000000	0.390513000	78.97720	32.772900
Max	171.732000	0.79984600	0.841339000	0.900000000	264.77500	83.954200
Result_mean	144.187556	0.60543766	0.104220112	0.895480466	199.30387	65.216003

Result_sd	6.783018	0.06951958	0.008331541	0.006511257	17.05541	4.506849
	rOM_ClxHorse.1.	f_Slope.1.	Sigma_W.1.			
Result_mode	0.0000133431	1.8609000	0.31043900			
2.5%	0.0000179869	0.9905980	0.25706400			
97.5%	0.0090747505	3.0518085	0.35981200			
Min	0.0000111440	0.7019950	0.21386700			
Max	0.0186332000	4.7352400	5.24867000			
Result_mean	0.0036650055	1.7929604	0.30592108			
Result_sd	0.0027609643	0.5178161	0.04682376			

Number of observations, n = 277
 Number of parameters, k = 8
 Log-likelihood, LL = -144.6308
 BIC = 334.2537

Potential scale reduction factors:

	Point est.	Upper C.I.
E_coc.1.	1.01	1.01
Ehp.1.	1.01	1.04
f_Slope.1.	1.01	1.04
Fm.1.	1.01	1.01
kap.1.	1.01	1.02
pAm.1.	1.01	1.02
r_pAm_pM.1.	1.00	1.00
rOM_ClxHorse.1.	1.00	1.01
Sigma_W.1.	1.00	1.01

Multivariate psrf

1.02

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)

```

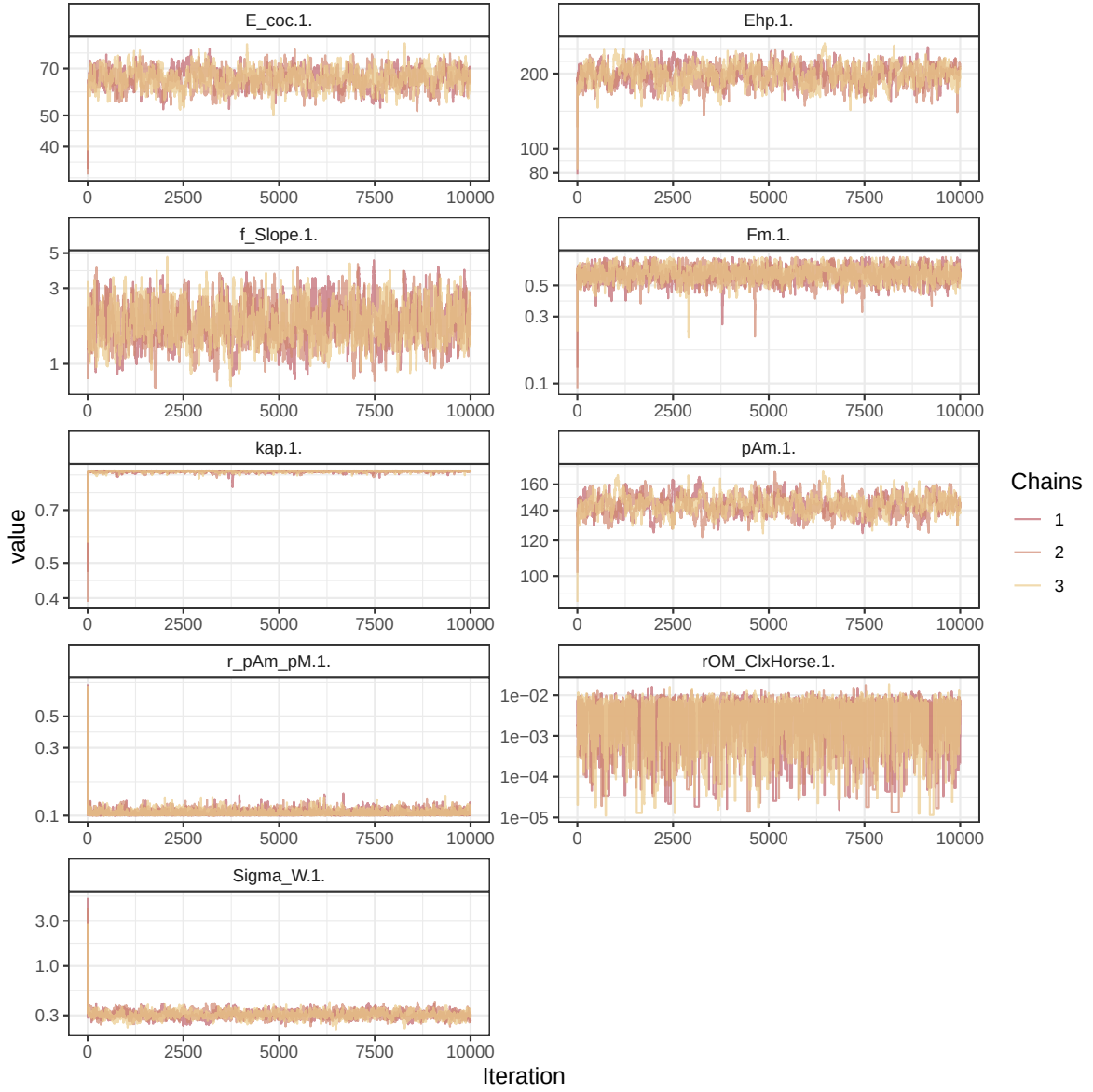


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

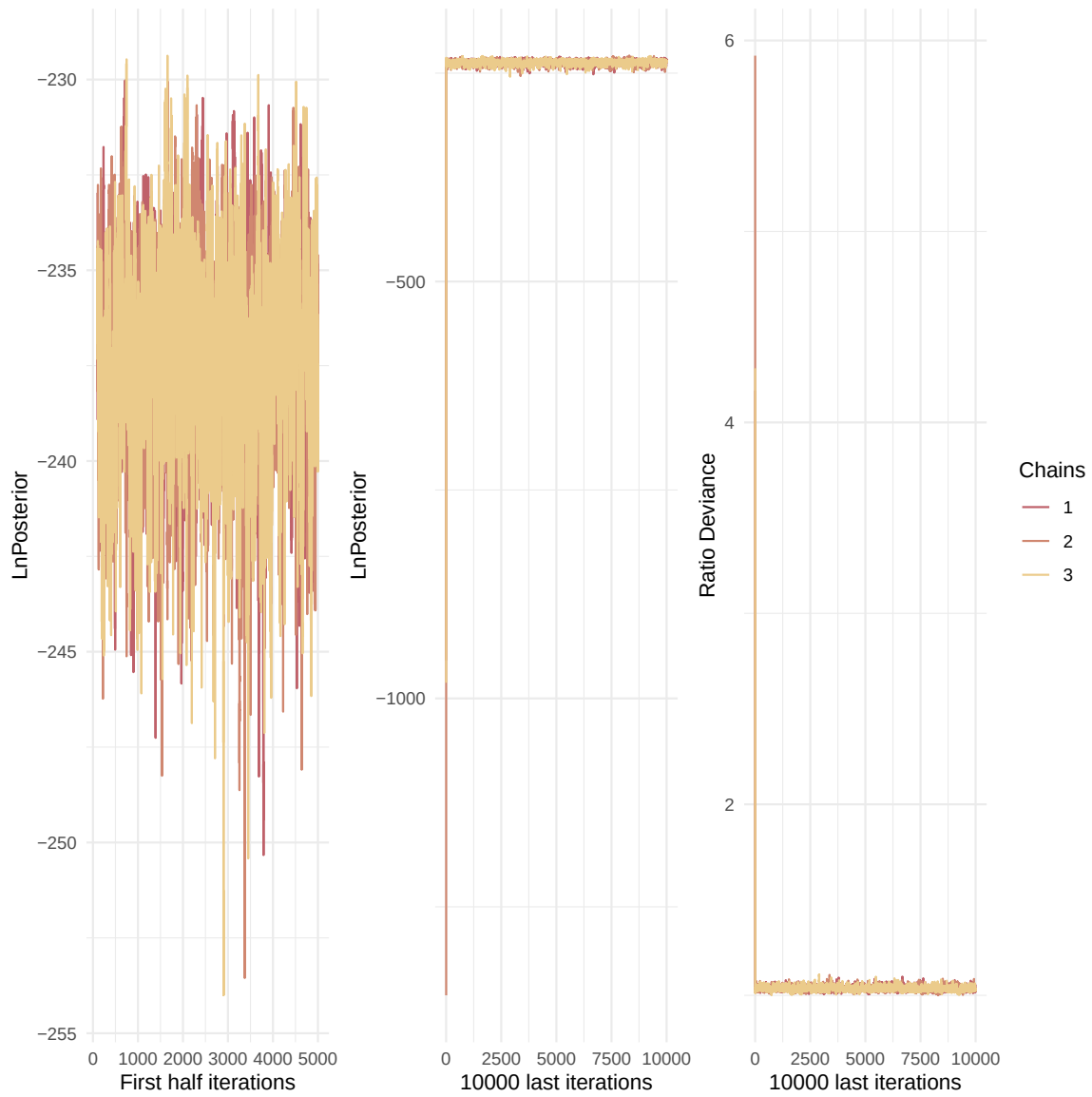


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

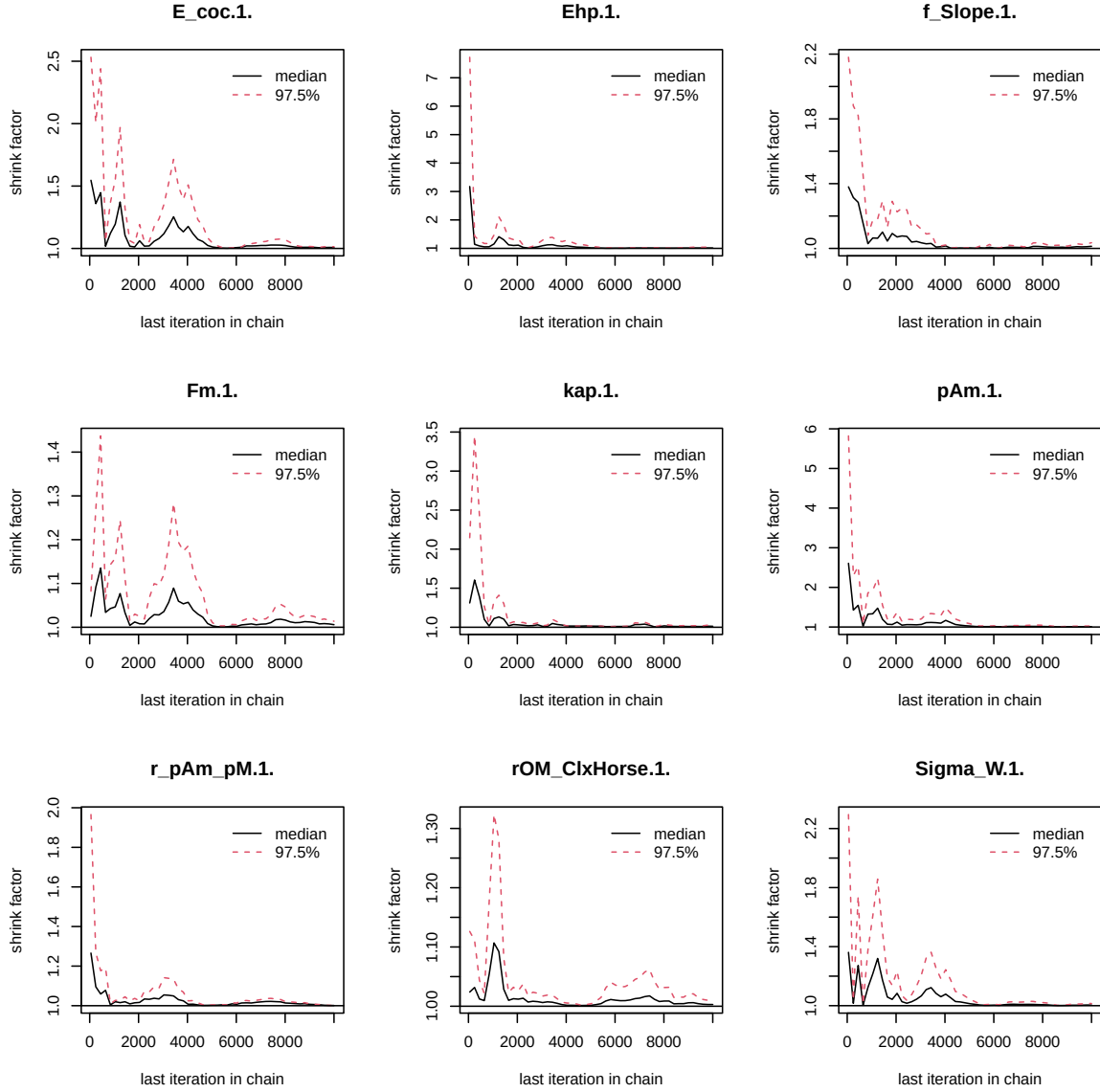


Figure 3: Chains convergence.

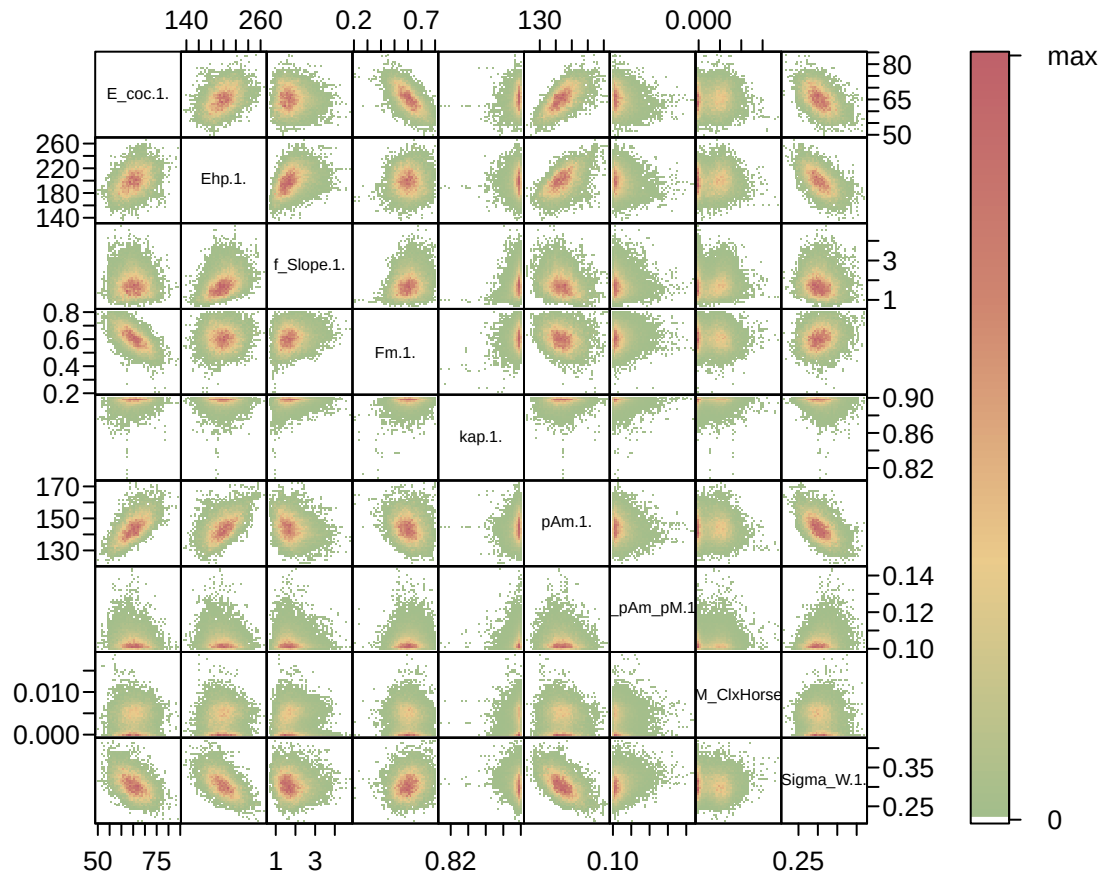


Figure 4: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

Posterior
and prior distributions

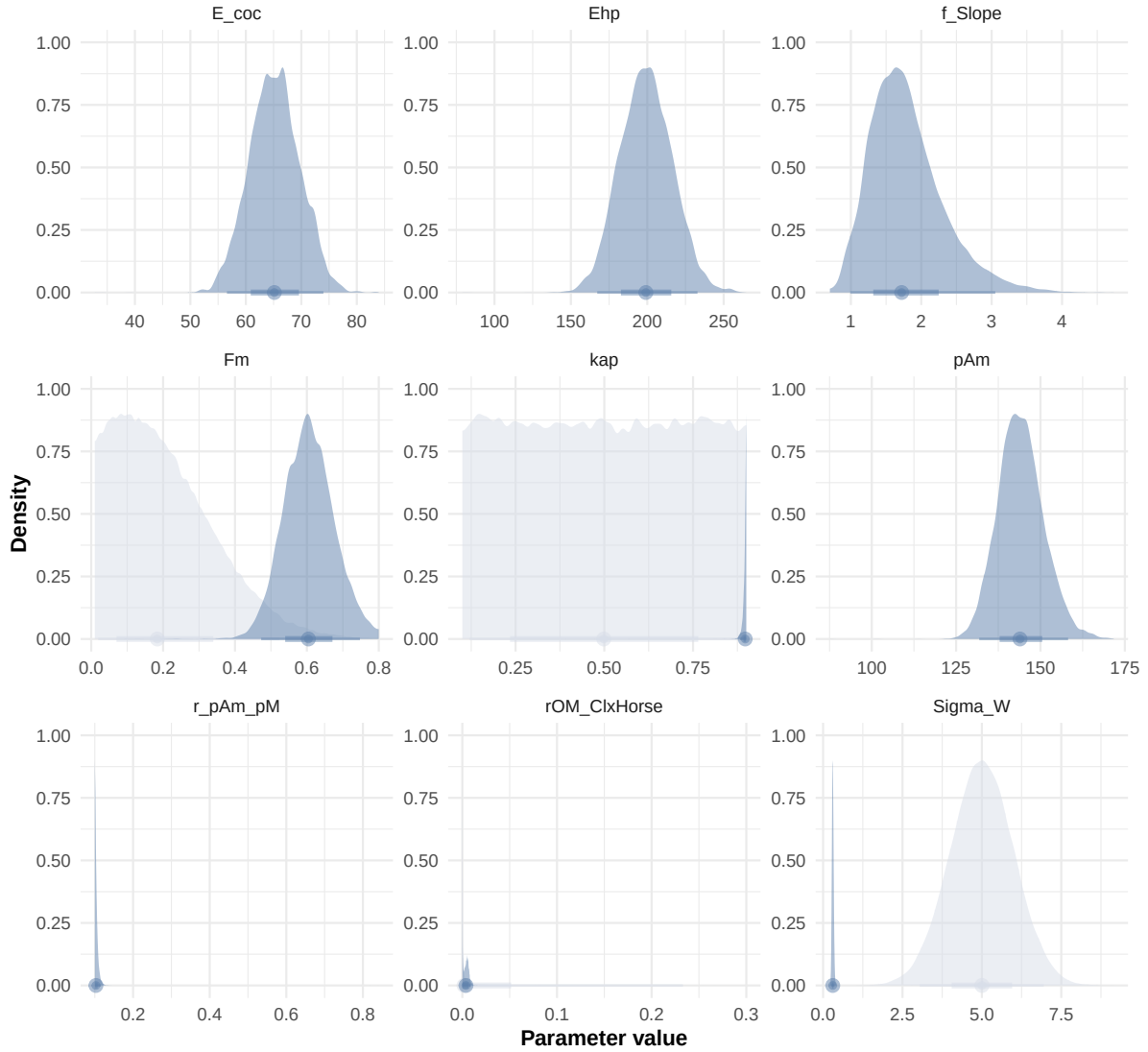


Figure 5: Distributions of the priors and the posteriors.


```
"Bart2020",  
#"Gollot2026_lufa"  
"Gollot2026_dens_D1",  
"Gollot2026_dens_D2",  
"Gollot2026_dens_D2b",  
"Gollot2026_dens_D5",  
"Gollot2026_dens_D7"  
)  
  
Adults_alone <- FALSE  
  
df_data <- f_import_data_DEB(l_Experiments, Adults_alone)  
  
text_param <- Summary_res["Result_mode", , drop = FALSE] %>%  
  t() %>%  
  as.data.frame() %>%  
  tibble::rownames_to_column("param") %>%  
  rename(val = 2) %>%  
  mutate(  
    param = gsub("\\\\.1\\.$|\\.1\\.\"", "", param),  
    line = glue("{param} = {val};")  
  ) %>%  
  pull(line) %>%  
  paste(collapse = "\n")  
  
Endpoints_print <- "Weight,  
  Length_struct,  
  Reproduction,  
  Energy,  
  Organic_matter,"  
  
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)  
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr  
# ./mcsim.DEBcaliSWI DEB_sim_full.in  
  
#####  
df_No_sim <- df_data |>  
  dplyr::select(ID_experiment, No_sim)  
  
df_carac_color <- data.frame(  
  carac = unique(l_carac),
```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)

```

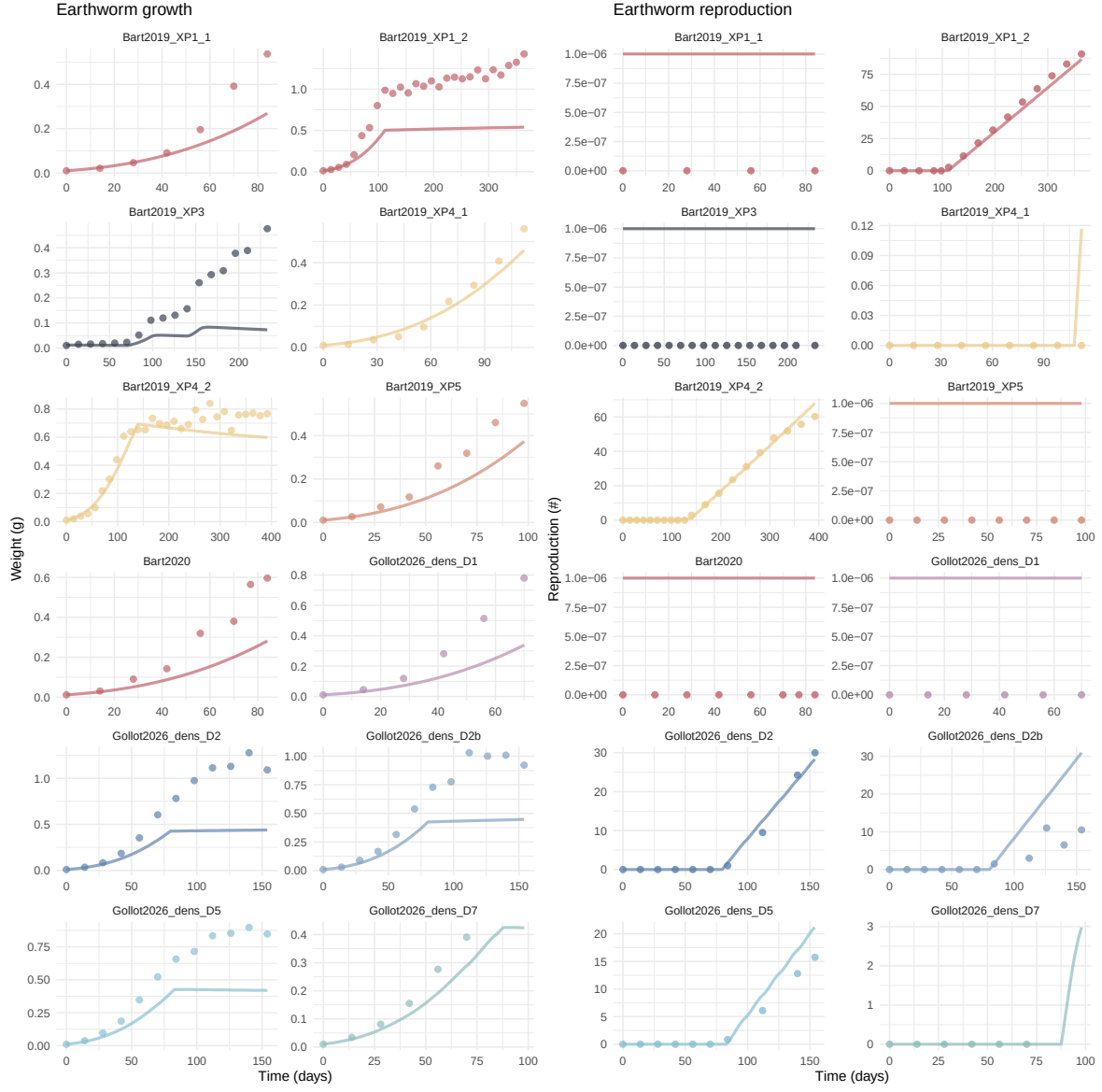
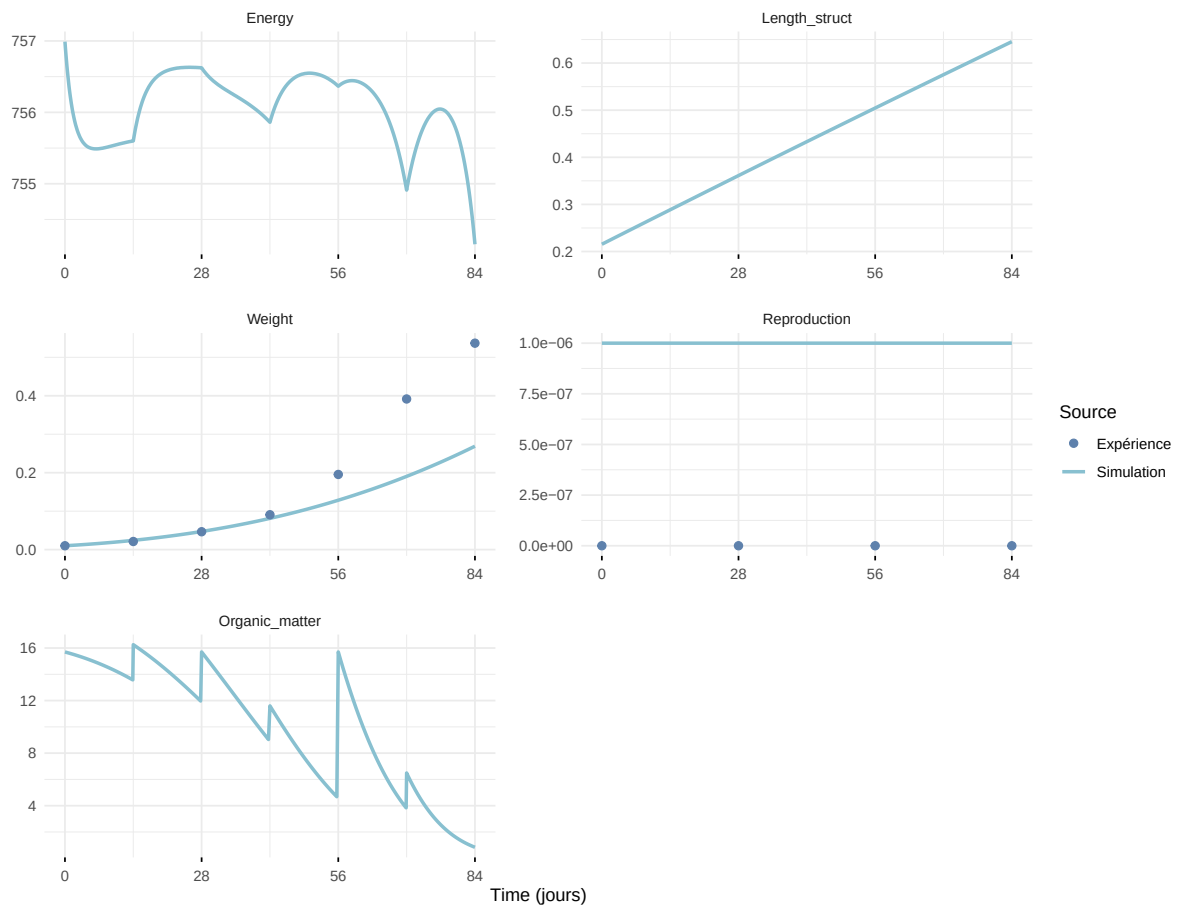
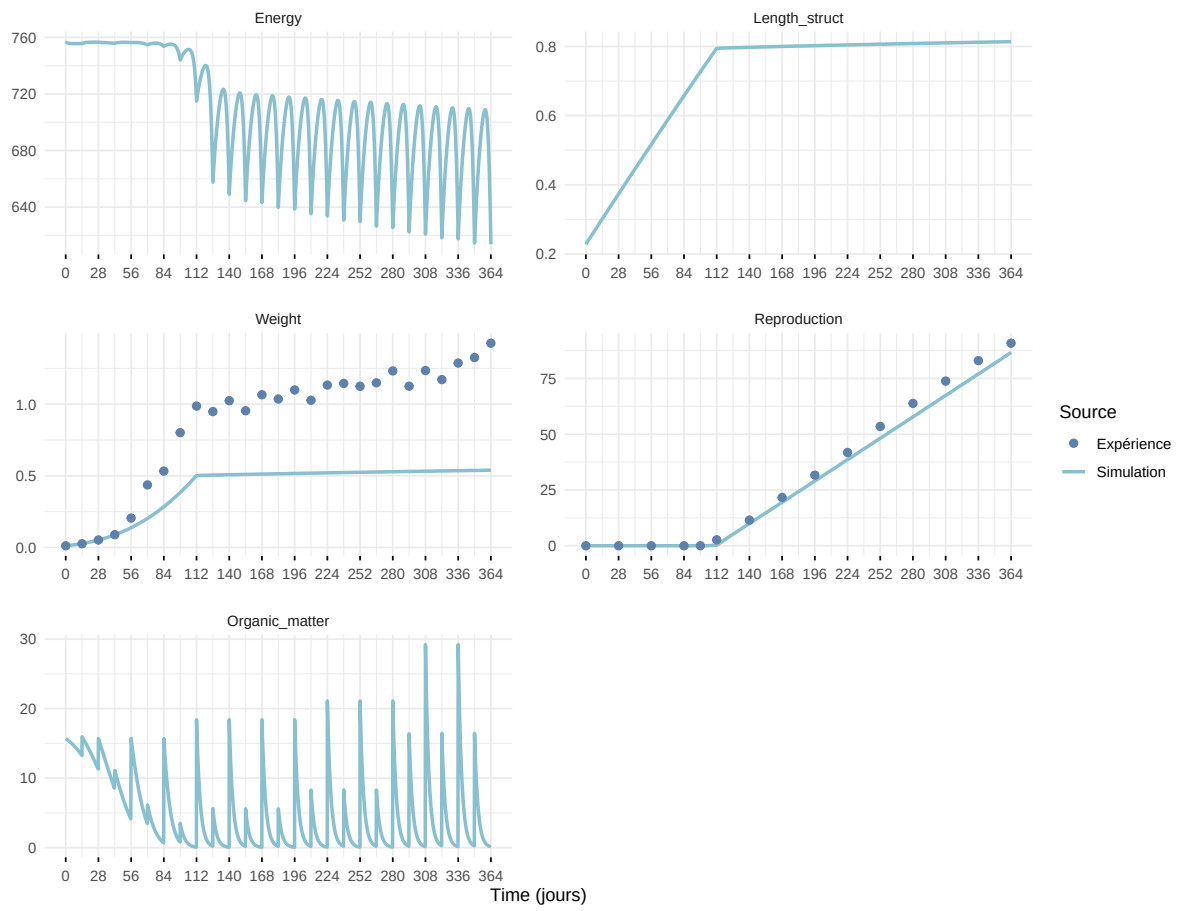


Figure 6: Simulations Weight and Reproduction.

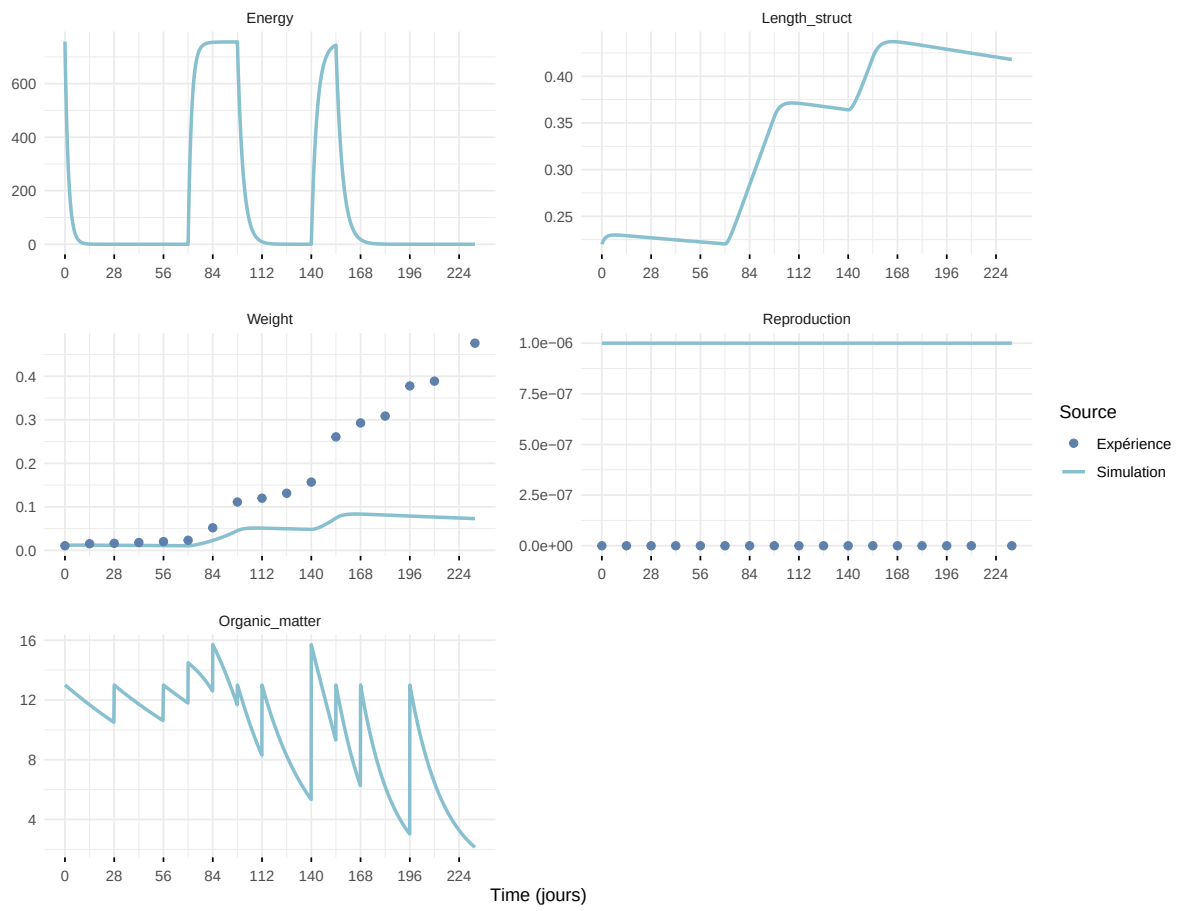
Expérience : Bart2019_XP1_1



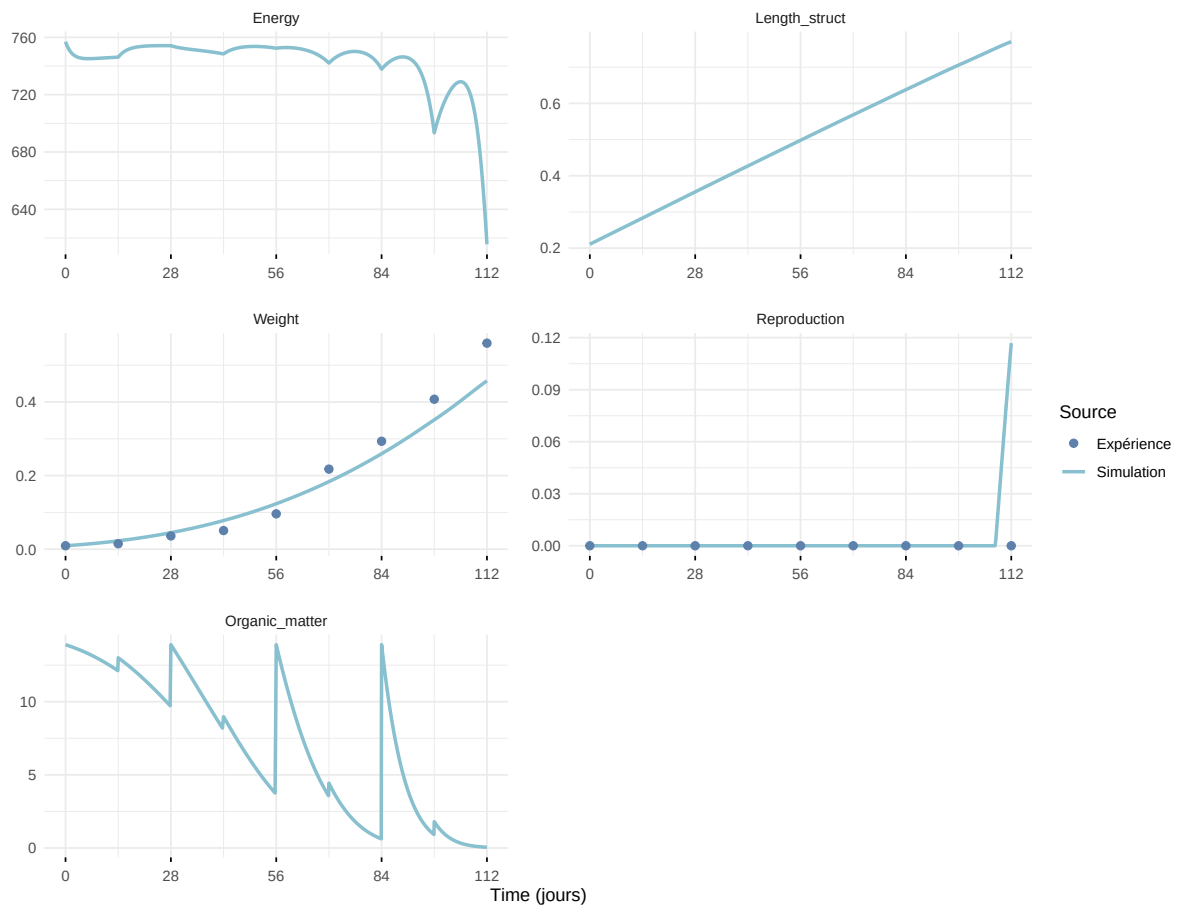
Expérience : Bart2019_XP1_2



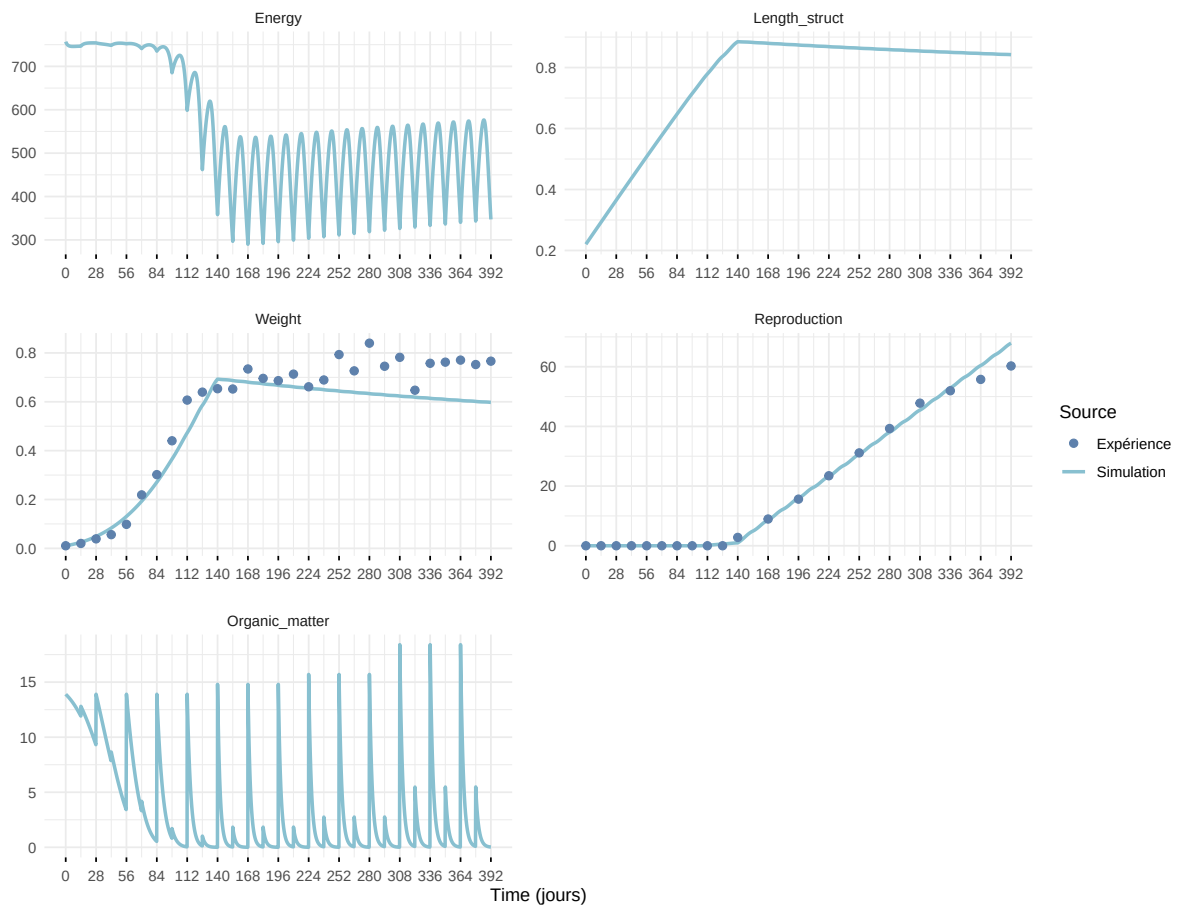
Expérience : Bart2019_XP3



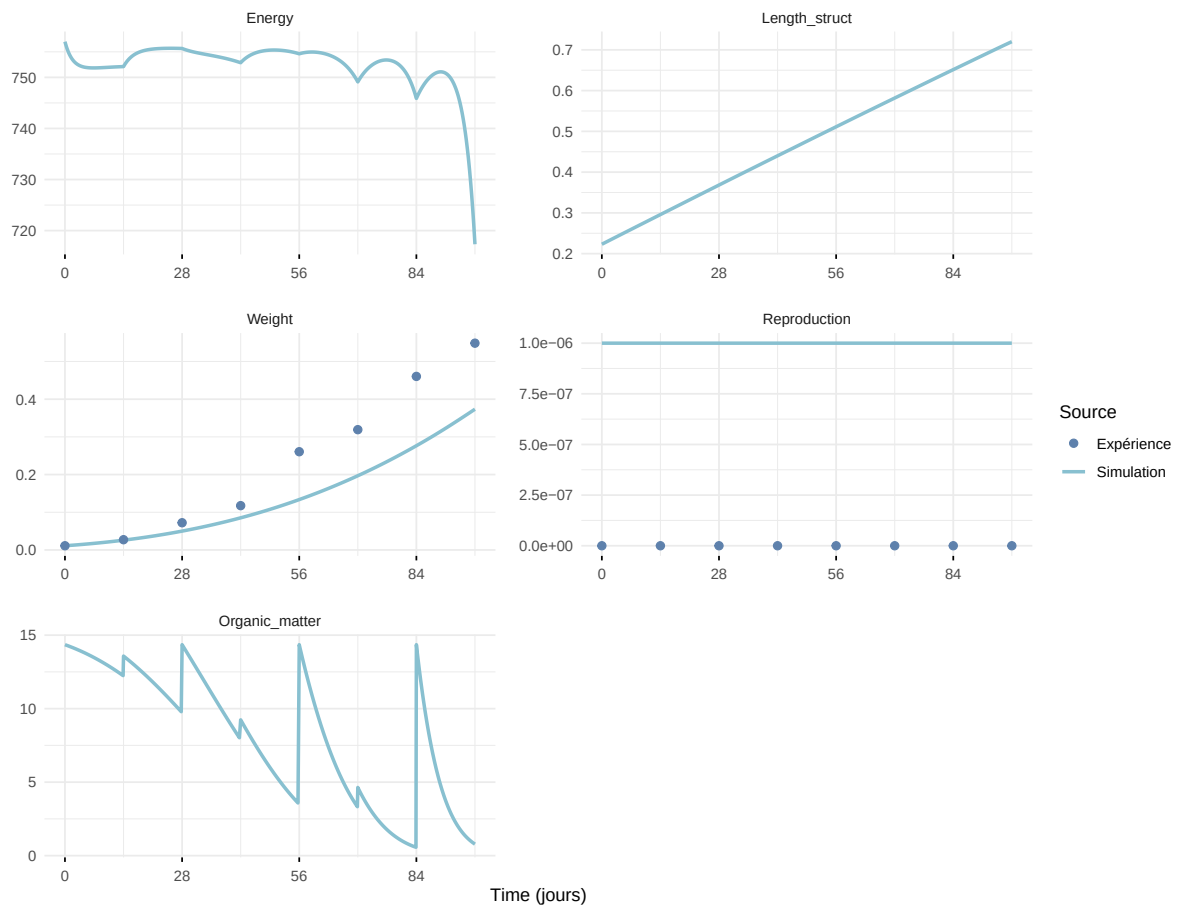
Expérience : Bart2019_XP4_1



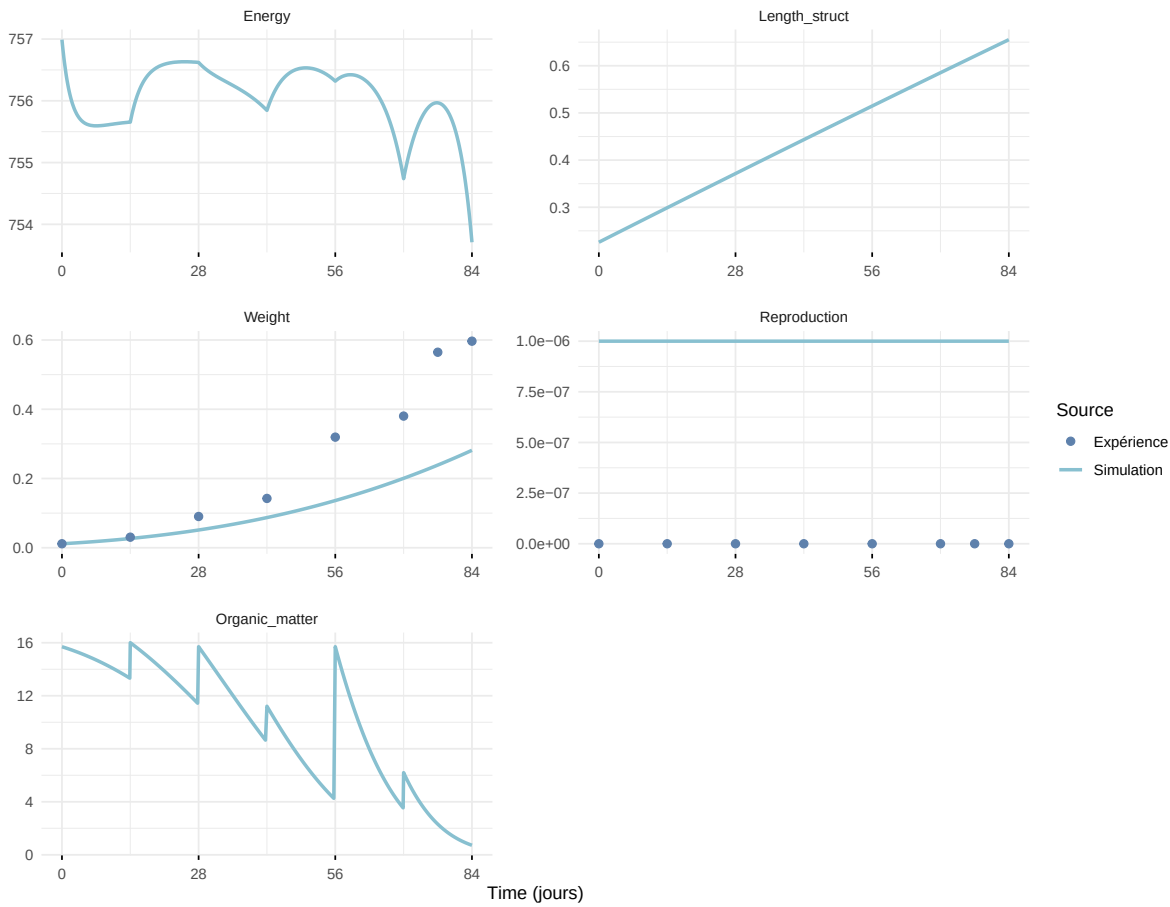
Expérience : Bart2019_XP4_2



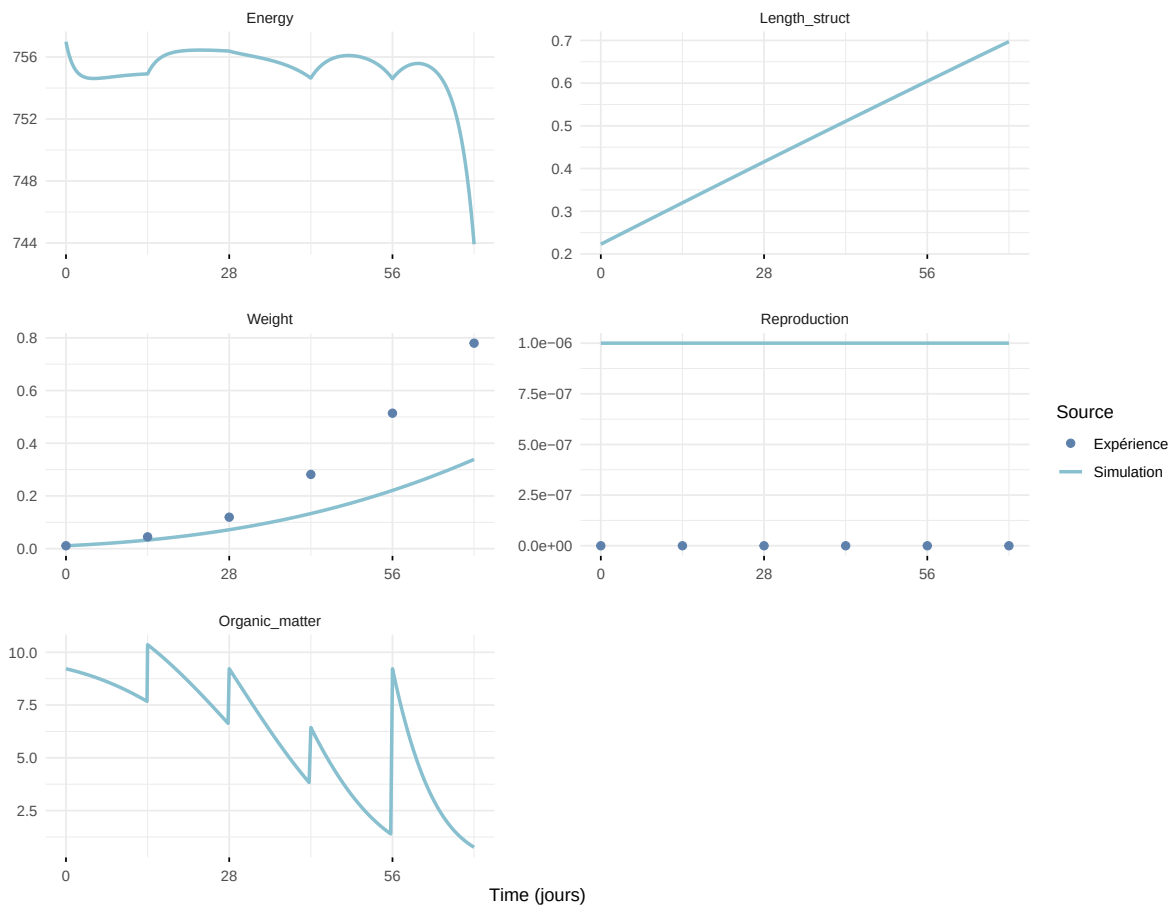
Expérience : Bart2019_XP5



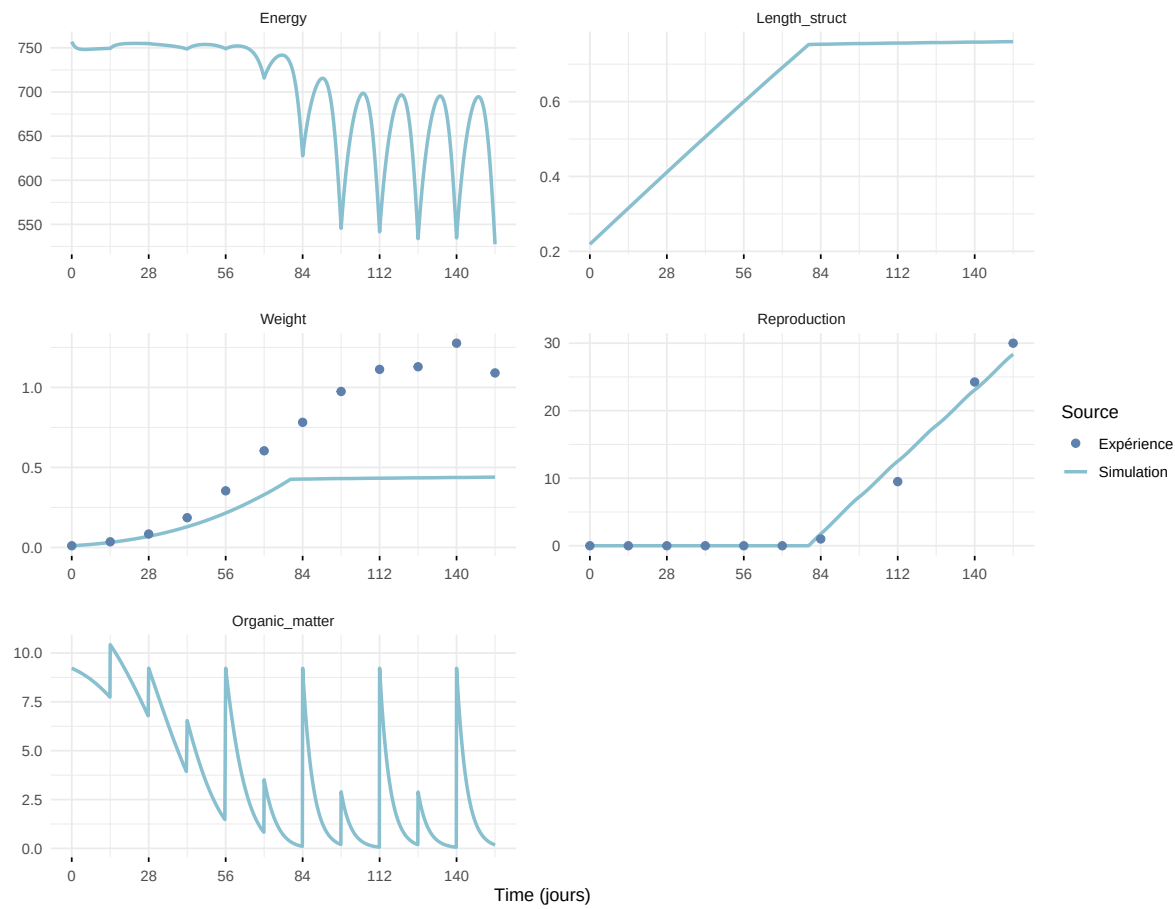
Expérience : Bart2020



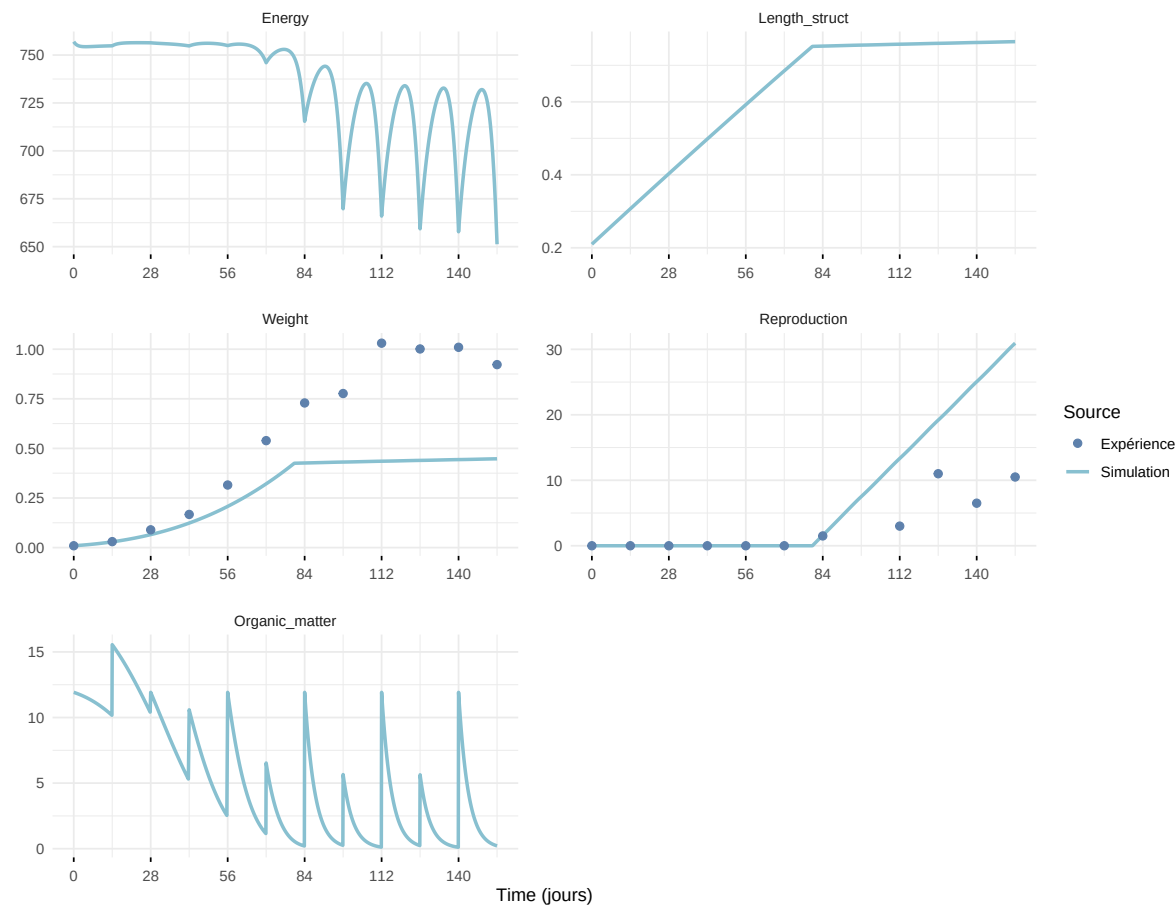
Expérience : Gollot2026_dens_D1



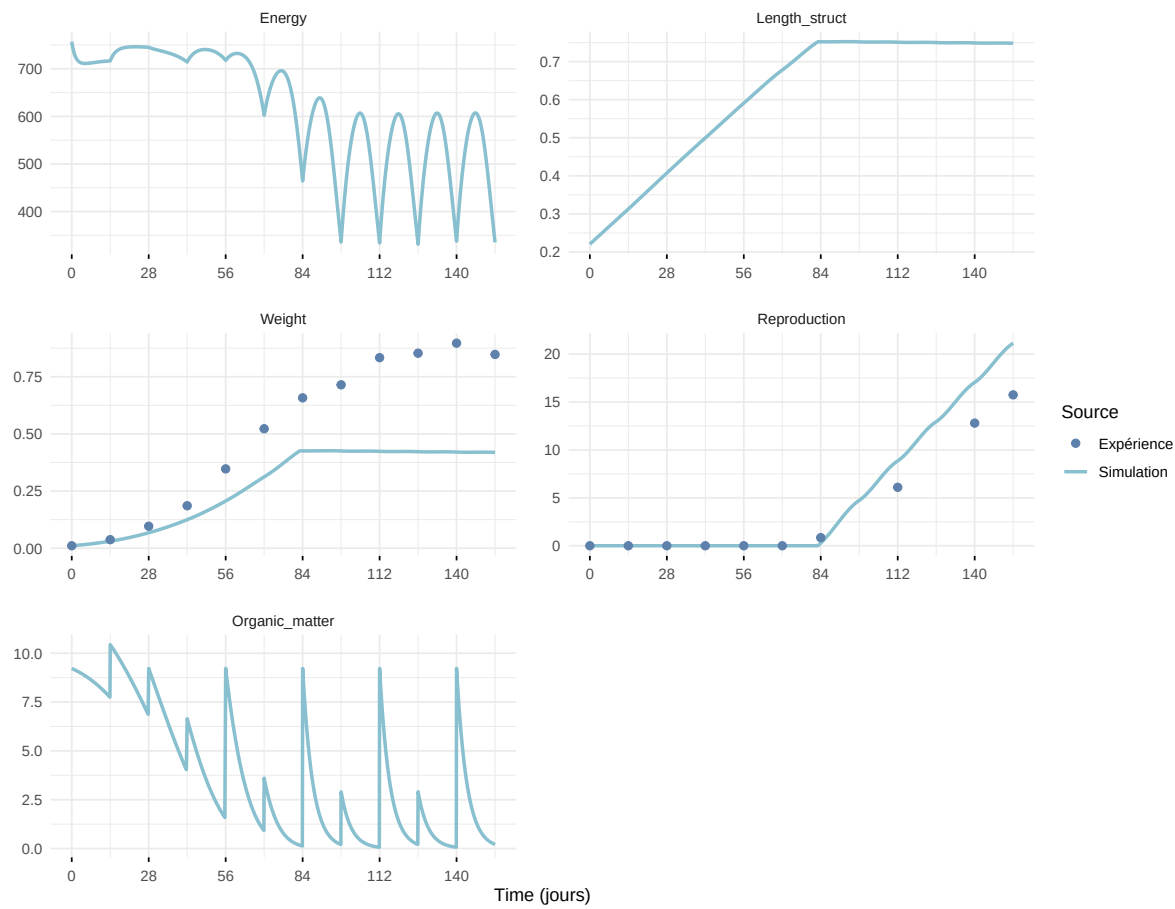
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

